Limitations of PCA

- **Linear Assumption:** PCA assumes that relationships between features are linear, which may limit its effectiveness for non-linear datasets.
- **Interpretability:** The transformed features (principal components) may not have a clear interpretation, making it harder to explain results.
- **Scaling Dependence:** Proper standardization is crucial, as unscaled features can distort the principal components.

Implementation in Python

The following example demonstrates PCA in Python using Scikit-Learn:

Step 1: Importing Libraries

```
from sklearn.decomposition import PCA from sklearn.preprocessing import StandardScaler
```

Step 2: Example Dataset

We create a simple dataset to illustrate PCA: # Example data

```
features = [
    [1.0, 1.0],
    [1.5, 2.0],
    [3.0, 4.0],
    [5.0, 7.0],
    [3.5, 5.0],
    [4.5, 5.0],
    [3.5, 4.5]

                ]
```

Step 3: Standardization

Standardize the features to ensure equal contribution to the principal components: scaler = StandardScaler()

```
scaled_features = scaler.fit_transform(features)
```

Step 4: Applying PCA

Reduce the data to two dimensions: pca = PCA(n_components=2)

```
principal_components = pca.fit_transform(scaled_features) print(principal_components)
```